RODRIGO ALARCON

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Summary

Aerospace engineer with an M.S. in Aerospace Engineering, focused on flight mechanics and controls, and B.S. degrees in Aerospace Engineering and Computer Science, seeking a Guidance, Navigation and Controls role. Designs, simulates and analyzes guidance and control systems, from equations of motion, system identification and state estimation to model predictive, nonlinear and adaptive control, autonomous motion planning and orbital guidance, in MATLAB/Simulink and Python. Currently a Quality Engineer on legacy avionics, driving root-cause investigations and working directly with customers and suppliers. Eager to keep growing and learning new tools and skills, both out of genuine interest and to expand the value brought to every team. Open to relocation and remote work.

Education

Master of Science: **Aerospace Engineering** **08/2026**
Florida Institute of Technology | Focus: Flight Mechanics and Controls | GPA: 3.6 Melbourne, FL

Bachelor of Science: **Aerospace Engineering** **05/2025**
Florida Institute of Technology Melbourne, FL

Bachelor of Science: **Computer Science** **05/2025**
Florida Institute of Technology Melbourne, FL

Representative Coursework

Guidance and Navigation of Aerospace Vehicles, Nonlinear Control Systems, Multivariable Feedback Control, Control Systems, Space Vehicle Control, Spaceflight Mechanics and Controls, Intelligent Controls, Aerodynamics and Flight Mechanics, Advanced Dynamics, Modeling Dynamic Systems, Robotics, Nonlinear Optimization, Deep Learning

Experience

Quality Engineer **12/2025 to Present**
Extant Aerospace (TransDigm), Airborne Electronics **Melbourne, FL**

- Drove root cause and corrective action (RCCA) efforts using 8D and 5 Whys on a long-standing issue with an inherited avionics product line, partnering directly with the customer to align expectations and drive it to resolution.
- Serve as a quality point of contact for customers and suppliers across the U.S. and abroad, resolving customer concerns and aligning on requirements.
- Investigate nonconformances and customer quality notices across multiple product lines and present dispositions at the Material Review Board (MRB).
- Author AS9102 Rev C first article inspection reports (FAIRs) for new production builds in an AS9100 environment, using DISCUS and Net-Inspect to verify conformance to drawings and specifications.
- Develop test procedures for avionics units and maintain accurate, current work instructions.
- Propose and approve engineering change orders affecting product design, test and production.
- Verify test data and unit build records before shipment, and review contracts and purchase orders to capture and flow down quality requirements.
- Prepare units for, and support, customer source inspections and acceptance.

Tutor **10/2026 to Present**
Grade Potential Tutoring **Melbourne, FL**

- Tutor students in Differential Equations and computer skills, adapting explanations to each student's level.

Controls Lead, Aerospace Senior Design **08/2023 to 04/2024**
Hydrofoil, Florida Tech **Melbourne, FL**

- Led control system design for a retractable hydrofoil attachment on a remote-controlled aircraft, implementing extension, retraction and active pitch control with a Pixhawk and ArduPilot autopilot.
- Planned and executed flight tests, monitored and adjusted flight systems in real time from a ground station, and turned flight-log analysis into design iterations.
- Overhauled the aircraft's flight components, replacing outdated hardware to improve performance and reliability.
- Facilitated design reviews with mechanical and aerodynamics leads, translating requirements into testable specifications.

Undergraduate Researcher **05/2023 to 09/2023**
Space Lab, Florida Tech **Melbourne, FL**

- Modeled and simulated a passive layered system for collecting non-trackable debris in Low Earth Orbit, using astrodynamics simulation to optimize orbital parameters for collection efficiency.
- Co-authored: Paula do Vale Pereira, Joao L. Silva Cotta, Jacob Mesley and **Rodrigo Alarcon**. "Preliminary Modeling of Not-trackable Debris Collection Using a Passive Layered System," AIAA 2023-4629. ASCEND 2023.

Projects

Cart-Pole System Identification and Model Predictive Control **2026**
MATLAB, Python, MuJoCo, PyTorch

- Identified nonlinear cart-pole dynamics with DMDc, SINDYc and neural networks, and used the identified model in a linear MPC that stabilized the nonlinear plant from every tested initial state.
- Compared MPC, MPPI, reinforcement learning and imitation learning (behavior cloning, DAgger) controllers in MATLAB and MuJoCo.
- Trained a neural network to approximate the MPC controller at a fraction of its computation time, and tuned MPC weights by constrained optimization to lower cost while eliminating constraint violations.

Multirotor MPC Motion Planning Through Point-Cloud Obstacles **2026**
Python, CVXPY

- Reproduced a published 3D planner: DBSCAN clustering, Voronoi/Dijkstra global routing, cylindrical flight corridors and receding-horizon convex MPC, guiding a multirotor collision-free through a 3D tunnel to its goal.

Lunar Lander Nonlinear and Adaptive Control **2025**
MATLAB

- Derived 3D lander dynamics and designed a RISE tracking controller with Lyapunov stability analysis under unknown mass, gravity and disturbances, driving the lander onto its descent trajectory from a large initial offset.

Spaceflight Mechanics and Sequential Rendezvous **2025**
Python

- Built orbit propagation with gravity-harmonic, drag and lunar perturbations, Lambert targeting with finite burns, and Clohessy–Wiltshire relative-motion feedback; the chaser completed rendezvous with its first target.

Skills

- Control Systems Engineering
- Model Predictive Control
- Nonlinear and Adaptive Control
- Autonomous Motion Planning
- Robot Kinematics and Dynamics (DH, Lagrange)
- Kalman Filtering and State Estimation
- System Identification (DMDc, SINDy)
- State-Space Modeling
- 3-DOF Vehicle Simulation
- Monte Carlo Analysis
- Astrodynamics Simulation
- Coordinate Frames and Quaternions
- Neural Networks and CNNs (U-Net)
- Reinforcement and Imitation Learning
- Flight Test Planning and Log Analysis
- Root Cause Analysis (8D, 5 Whys)
- MATLAB
- Simulink
- Python (NumPy, SciPy, CVXPY)
- PyTorch
- scikit-learn
- C++
- CMake
- Pixhawk and ArduPilot Autopilots
- ROS 2 (C++, Python)
- OpenCV
- MuJoCo
- Git
- Linux
- Creo Parametric
- Arduino Electronics
- Electronics Troubleshooting
- AS9100 / AS9102 Rev C FAI
- Epicor, Net-Inspect, DISCUS
- MS Office (Excel, Word, PowerPoint)

Honors and Certifications

Private Pilot Certificate, FAA 2020
Eagle Scout, Boy Scouts of America 2018